## Recap - Set

```
s = set()
len(s)
```

```
0
```

```
type(s)
```

```
set
```

```
s = {3,5,7,3,7,8,9,1}
len(s)
```

```
6
```

```
print(s)
```

```
{1, 3, 5, 7, 8, 9}
```

```
s.add(10)
print(s)
```

```
{1, 3, 5, 7, 8, 9, 10}
```

```
s.remove(7)
```

```
print(s)
```

```
{1, 3, 5, 8, 9, 10}
```

```
s.pop()
```

```
1
```

```
print(s)
```

```
{3, 5, 8, 9, 10}
```

```
3 in s
```

```
True
```

```
5 in s
```

```
True
```

```
4 in s
```

```
False
```

```
a = [4,6,7,3,4,8,7]
len(a)
```

```
7
```

```
s = set(a)
print(s)
```

```
{3, 4, 6, 7, 8}
```

```
len(s)
```

```
5
```

```
len(a)
```

```
7
```

```
s1 = {3,4,6}
s2 = {1,3,7}
s1.union(s2)
```

```
{1, 3, 4, 6, 7}
```

```
s1 | s2
```

```
{1, 3, 4, 6, 7}
```

```
s1.intersection(s2)
```

```
{3}
```

```
s1 & s2
```

```
{3}
```

```
s1 - s2
```

```
{4, 6}
```

```
s2 - s1
```

```
{1, 7}
```

```
s = set("Banana")
print(s)
```

```
{'B', 'n', 'a'}
```

## HW

```
words = ["apple", "banana", "apple", "mango", "banana"]

unique = set(words)

result = ""

for word in unique:
    result += word[0]

print(len(result))
```

```
3
```

```
text = "mississippi"

letters = set(text)

count = 0

for ch in letters:
    if ch in "aeiou":
        count += 1

print(count)
```

```
1
```

```
A = {10, 20, 30}

B = {30, 40, 50}
```

```
C = A.union(B)
print(C)
print(min(C), max(C))
```

```
{50, 20, 40, 10, 30}
10 50
```

```
numbers = [1, 2, 2, 3, 4, 4, 5]

unique = set(numbers)

even = set()

for num in unique:
    if num % 2 == 0:
        even.add(num)

print(len(even))
```

```
2
```

```
students = {
    "Rahul",
    "Amit",
    "Neha"
}

students.remove("Amit")

students.add("Riya")

print(len(students))
```

```
3
```

```
colors = {"Red", "Green", "Blue"}

result = set()

for color in colors:
    result.add(color.upper())

print(result)
print(len(result))
```

```
{'BLUE', 'RED', 'GREEN'}
3
```

```
numbers = [10, 20, 20, 30, 40, 40]

A = set(numbers)

B = {20, 50}

result = A - B

print(result)
print(sum(result))
```

```
{40, 10, 30}
80
```

## Coding Practice

### Problem 1 : Student Grade Analyzer

```
Problem Statement

Write a Python program that takes marks of 5 subjects as input.

Your program should:
```

```
Store the marks in a list.
Calculate the average marks.
Find the highest and lowest marks.
Display the grade using:

| Average  | Grade |
| -------- | ----- |
| 90+      | A     |
| 75-89    | B     |
| 60-74    | C     |
| Below 60 | D     |
```

```
80
95
70
85
90
```

```
(80+95+70+85+90)/5
```

```
84.0
```

```
# high = 95, low = 70
# grade = B
```

```
marks = []
for i in range(5):
  m = int(input())
  marks.append(m)
```

```
80
95
70
85
90
```

```
s = sum(marks)
n = len(marks)

avg = s/n
print(avg)
```

```
84.0
```

```
mn = min(marks)
mx = max(marks)
print(mn, mx)
```

```
70 95
```

```
| Average  | Grade |
| -------- | ----- |
| 90+      | A     |
| 75-89    | B     |
| 60-74    | C     |
| Below 60 | D     |
```

```
if avg >= 90:
  grade = "A"
elif avg >=75:
  grade = "B"
elif avg >= 60:
  grade = "C"
else:
  grade = "D"

print(grade)
```

```
B
```

```
#final systamtic code:

marks = []

for i in range(5):
    mark = int(input())
    marks.append(mark)

average = sum(marks) / len(marks)

print("Average :", average)
print("Highest :", max(marks))
print("Lowest :", min(marks))

if average >= 90:
    print("Grade : A")
elif average >= 75:
    print("Grade : B")
elif average >= 60:
    print("Grade : C")
else:
    print("Grade : D")
```

## Problem 2 : Word Frequency Counter

```
Problem Statement

Take a sentence as input.

Display how many times each word appears.


Eg1: python is fun python is easy
op: python - 2, is - 2 , easy - 1, fun - 1
```

```
s = input()
words = s.split(" ")
print(words)
```

```
python is fun python is easy
['python', 'is', 'fun', 'python', 'is', 'easy']
```

```
freq = {}
for word in words:
  if word not in freq:
    freq[word] = 1
  else:
    freq[word] += 1

print(freq)
```

```
{'python': 2, 'is': 2, 'fun': 1, 'easy': 1}
```

## Problem 3 : Remove Duplicate Names

```
Problem Statement

Take names separated by spaces.

Remove duplicate names while preserving the first occurrence.

Eg1: Rahul Amit Rahul Neha Amit
op: Rahul Amit Neha
```

```
s = input()
names = s.split()
```

```
Rahul Amit Rahul Neha Amit
```

```
print(names)
```

```
['Rahul', 'Amit', 'Rahul', 'Neha', 'Amit']
```

```
unique_names = []
for name in names:
  if name not in unique_names:
    unique_names.append(name)

print(unique_names)
```

```
['Rahul', 'Amit', 'Neha']
```

```
for name in unique_names:
  print(name)
```

```
Rahul
Amit
Neha
```

```
#well formatted code:

names = input().split()

unique = []

for name in names:
    if name not in unique:
        unique.append(name)

for name in unique:
    print(name)
```

```
Aditya Shyam Prshant Aditya
Aditya
Shyam
Prshant
```

## Problem 4 : Character Counter

```
Problem Statement

Take a string as input from the user.

Display

Number of vowels
Number of consonants
Number of digits
Number of spaces
```

```
# Ip1: Python3 is Fun
# op1: vowels - 3
# consonants - 8
# digits - 1
# spaces - 2
```

```
s = input()
sl = s.lower()

vowels = 0
consonants = 0
digits = 0
spaces = 0

for c in sl:
  if c in "aeiou":
    vowels += 1
  elif c >= "a" and c <= "z":
    consonants += 1
```

```
  elif c >="0" and c <= "9":
    digits += 1
  elif c == " ":
    spaces += 1

print("Vowels -", vowels)
print("Consonants -", consonants)
print("Digits -", digits)
print("Spaces -", spaces)
```

```
Python3 is fun
Vowels - 3
Consonants - 8
Digits - 1
Spaces - 2
```

```
#version2

text = input()

vowels = 0
consonants = 0
digits = 0
spaces = 0

for ch in text:

    if ch.lower() in "aeiou":
        vowels += 1

    elif ch.isalpha():
        consonants += 1

    elif ch.isdigit():
        digits += 1

    elif ch == " ":
        spaces += 1

print("Vowels :", vowels)
print("Consonants :", consonants)
print("Digits :", digits)
print("Spaces :", spaces)
```

```
"A".isalpha()
```

```
True
```

```
"9".isalpha()
```

```
False
```

```
"b".isalpha()
```

```
True
```

```
"b".isdigit()
```

```
False
```

```
"9".isdigit()
```

```
True
```

## Problem 5 : Shopping Bill

```
Start coding or generate with AI.
```

```
Problem Statement
```

```
Take item names and prices for 5 products.

Store

item names in one list
prices in another list

Display

total bill amount
highest priced item
lowest priced item
```

```
# Eg1:
# Apple
# 100

# Milk
# 50

# Bread
# 40

# Rice
# 80

# Juice
# 120

# op1: 390
# Juice
# Bread
```

Double-click (or enter) to edit

```
names = []
prices = []

for i in range(5):
  name = input()
  price = int(input())

  names.append(name)
  prices.append(price)

print(names)
print(prices)
```

```
Apple
100
Milk
50
Bread
40
Rice
80
Juice
120
['Apple', 'Milk', 'Bread', 'Rice', 'Juice']
[100, 50, 40, 80, 120]
```

```
total_bill_amount = sum(prices)
mx = prices[0]
mx_prd = ""
mn = prices[0]
mn_prd = ""

for i in range(5):
  if prices[i] > mx:
    mx_prd = names[i]
    mx = prices[i]
  elif prices[i] < mn:
    mn_prd = names[i]
    mn = prices[i]
```

```
print(total_bill_amount)
print(mx, mx_prd)
print(mn, mn_prd)
```

```
390
120 Juice
40 Bread
```

## HW

```
Problem 6 : Password Validator
Problem Statement

A password is valid if it contains

minimum 8 characters
at least one uppercase letter
at least one lowercase letter
at least one digit

Display

Strong Password

or

Weak Password
```

```
Problem 7 : Cricket Scoreboard
Problem Statement

Take player names and runs.

Store them in a dictionary.

Display

Highest scorer
Average score
```

```
Start coding or generate with AI.
```

```
roblem 8 : Common Subjects
Problem Statement

Input two sets of subjects.

Display

Common subjects
Subjects only in first student
Subjects only in second student
```

```
Problem 9 : Most Frequent Character
Problem Statement

Find the character that appears the maximum number of times in a string.

Ignore spaces.
```

Double-click (or enter) to edit